Environmental Sciences (MC-401)

Course Details

Credit Points: 1

Pre-Requisites

1. Basic knowledge of Environmental Science

Syllabus Outline

Unit 1

Environment & Resources

- Basic concepts: man society and environment and their interrelationship.
- Mathematics of population growth and associated problems.
- Definition and types of resources (renewable, non-renewable, potentially renewable), effect of excessive use vis-à-vis population growth, Sustainable Development.
- Materials balance: steady state conservation system, steady state system with non-conservative pollutants, step function.
- Environmental degradation: natural hazards (flood, earthquake, landslide) and anthropogenic degradation (acid rain).
- Nature and scope of Environmental Science and Engineering.

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Unit 2

Ecology & Biodiversity

- Elements of ecology: system, open system, closed system, definition of ecology, species, population, community, definition of ecosystem — components, types and function.
- Forest, Grassland, Desert, Aquatic and Mangrove (Sundarban) ecosystems; food chains and food webs.

- Biogeochemical cycles (Oxygen, Carbon, Nitrogen, Phosphate, Sulphur).
- Biodiversity — types, importance, endemic species, biodiversity hot-spots, threats and conservation.

Unit 3

Atmosphere & Air Pollution

- Atmospheric composition: Troposphere, Stratosphere, Mesosphere, Thermosphere.
- Energy balance: conductive/convective and radiation heat transfer, simple global temperature model.
- Greenhouse effect, global warming and consequences.
- Lapse rates, atmospheric stability, temperature inversion.
- Atmospheric dispersion: Gaussian plume model.
- Pollutants — primary and secondary, criteria pollutants, emission standards.
- Sources and effects of SPM, CO_x, NO_x, SO_x, PAN.
- Smog, photochemical smog, London smog.
- Ozone layer depletion by CFCs.
- Air quality standards and control measures (ESP, cyclone separator, bag house, catalytic converter, scrubber).

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Unit 4

Water Pollution

- Hydrosphere, hydrological cycle and natural water.
- Pollutants of water — origin and effects: oxygen-demanding wastes, pathogens, nutrients, salts, thermal, heavy metals, pesticides, volatile organic compounds.

- River pollution: DO, BOD (5-day and seeded BOD test), BOD reaction rate constants, deoxygenation and reaeration, COD, oil, grease, pH.
- Lake eutrophication.
- Groundwater: aquifers, hydraulic gradient, groundwater flow.
- Water treatment systems (coagulation, flocculation, sedimentation, filtration, disinfection, softening).
- Wastewater treatment: primary, secondary (trickling filters, rotating biological contractor, activated sludge, oxidation ponds) and tertiary treatment.
- Toxic elements and biochemical effects: Lead, Mercury, Cadmium, Arsenic.

Unit 5

Solid Waste

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- Lithosphere; internal structure of earth, rock and soil.
- Municipal, industrial, commercial, agricultural, domestic, pathological and hazardous solid wastes.
- Recovery and disposal methods — open dumping, landfilling, incineration, composting, recycling.
- Solid waste management and control of hazardous and biomedical waste.

Unit 6

Noise Pollution

- Definition and effects of noise pollution.
- Classification: transport, occupational and neighbourhood noise.

- Noise frequency, noise pressure, noise intensity, threshold limit value, equivalent noise level, L10 (18hr Index), Ldn.
- Noise pollution control.

Unit 7

Environmental Management & Law

- Environmental Impact Assessment (EIA), Environmental Audit. Environmental laws and protection act of India.
- International environmental treaties, agreements and protocols.

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